

Cybersecurity

Windows Server Services

Lesson 7.4



Learning Objectives

By the end of today's class, you will be able to:

- 1 Understand Task Scheduling in a Windows Environment
- 2 Configure Windows Update on Windows Computers
- 3 Understand the Windows Attack Space



Welcome



Recap

Windows Administration and Hardening

Day 1

We used PowerShell to navigate and manage the Windows system directories and files.

Day 2

We secured our domain using group policy.

Day 3

We deployed Windows Defender to secure against malware threats.



Let's recap

Recap of previous class



We used Powershell to navigate and manage the Windows system directories and files



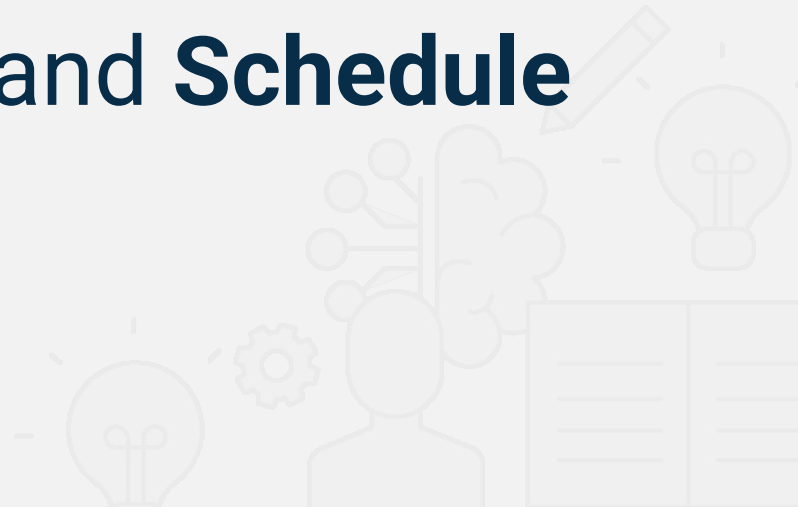
We secured our domain using group policy

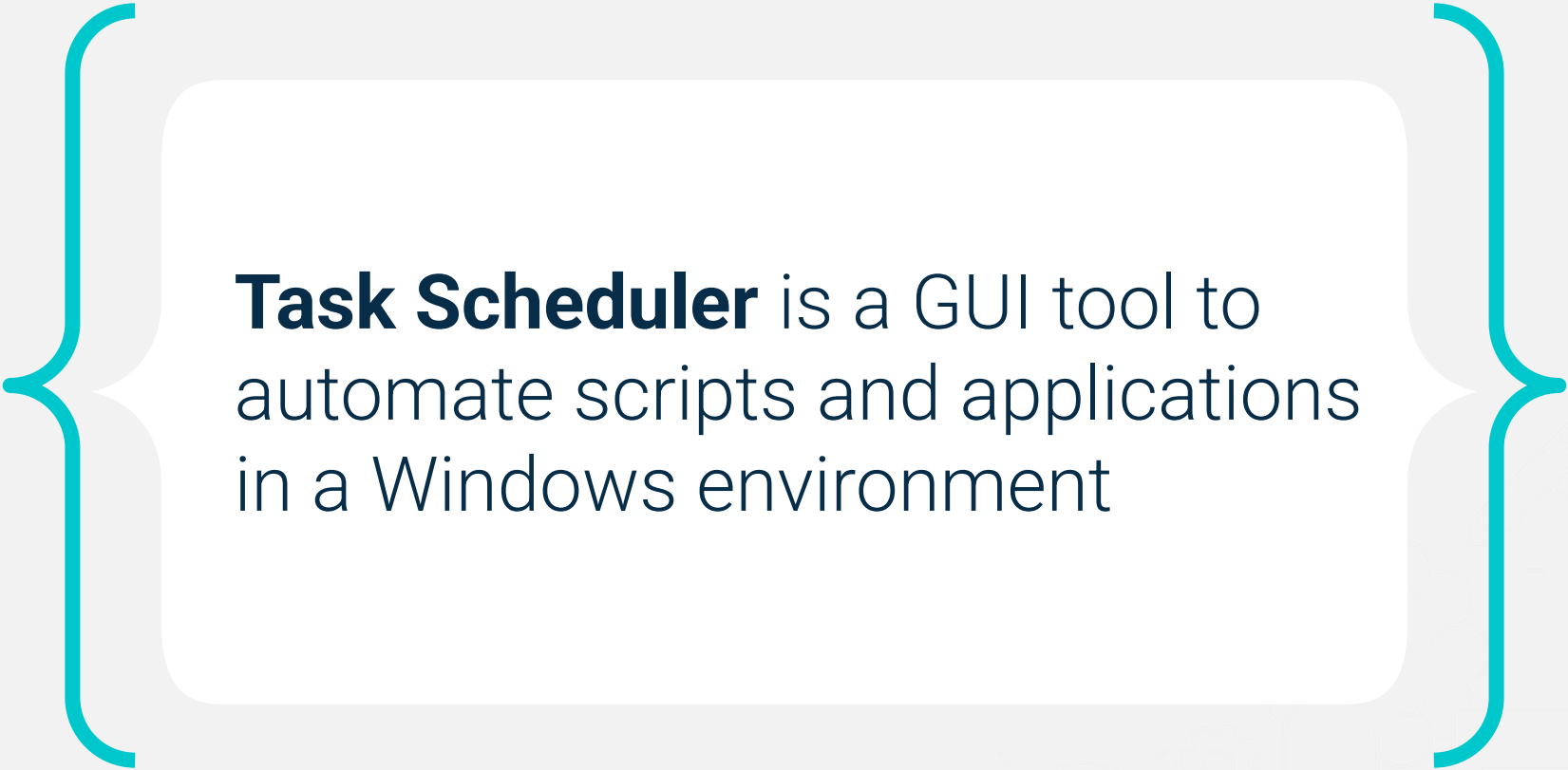


We deployed Windows Defender to secure against malware threats



Today, we're going to learn
how to **Protect Windows
Services and Schedule
Tasks.**





Task Scheduler is a GUI tool to automate scripts and applications in a Windows environment



Task Scheduler

FileActionViewHelp

Task Scheduler (Local)

Task Scheduler Library

Task Scheduler Summary (Last refreshed: 10/25/2019 1:49:41 PM)

Task Status

Status of tasks that have started in the following time period: Last 24 hours

Summary: 0 total - 0 running, 0 succeeded, 0 stopped, 0 failed

Task Name	Run Result	Run Start	Run End	Triggered By
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Active Tasks

Active tasks are tasks that are currently enabled and have not expired.

Summary: 95 total

Task Name	Next Run Time	Triggers	Location
Adobe Acrobat Update Task	10/25/2019 2:00:00 PM	Multiple triggers defined	\
GoogleUpdateTaskMachineCore	10/25/2019 2:02:26 PM	Multiple triggers defined	\
GoogleUpdateTaskMachineUA	10/25/2019 2:02:26 PM	At 2:02 PM every day - A...	\
Adobe Flash Player Updater	10/25/2019 2:17:00 PM	At 7:17 PM every day - A...	\
RefreshCache	10/25/2019 2:48:36 PM	Multiple triggers defined	\Microsoft\Windows\Fli...

Last refreshed at 10/25/2019 1:49:41 PM

Refresh

Actions

Task Scheduler (Local)

Connect to Another Computer...

Create Basic Task...

Create Task...

Import Task...

Display All Running Tasks

Enable All Tasks History

AT Service Account Configuration

View

Refresh

Help



Windows Tasks are
similar to cron jobs in
Linux



Scheduling Tasks

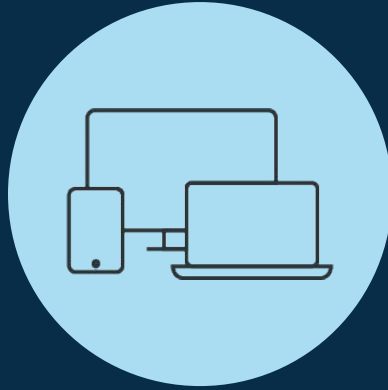
Imagine the following scenario:

Your CEO wants to schedule reports to be created daily.

The systems isn't remotely managed at this time.

We can use the Task Scheduler to create a task that will do this.





Instructor Demonstration

Task Scheduling



Activity:

Task Scheduling

Suggested Time:
15 Minutes





Time's up!
Let's review





Questions?

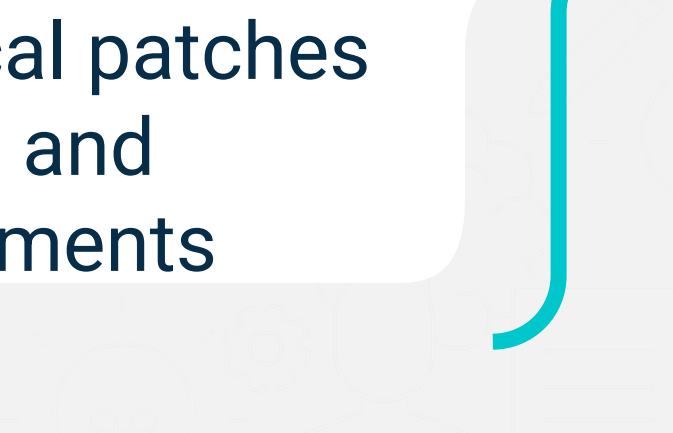


Windows Updates and Enterprise Management



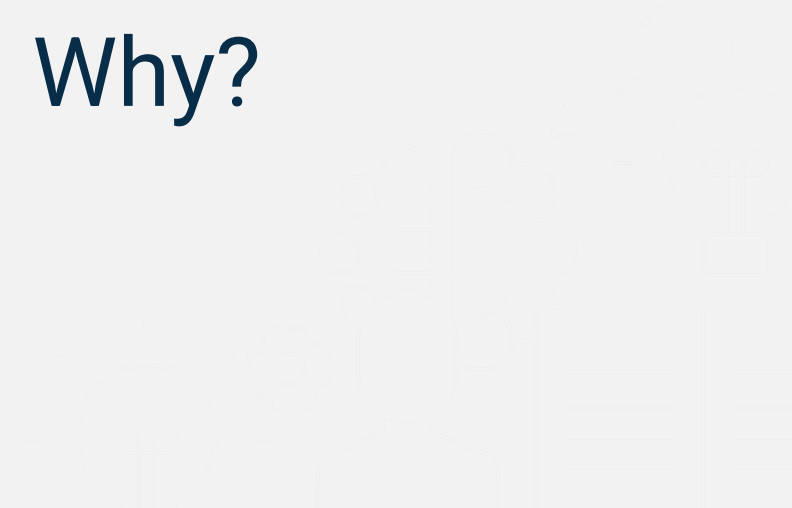
Updates are the backbone of a secure and reliable IT environment.

Updates provide critical patches against vulnerabilities and performance improvements





But... patches often go ignored. Why?





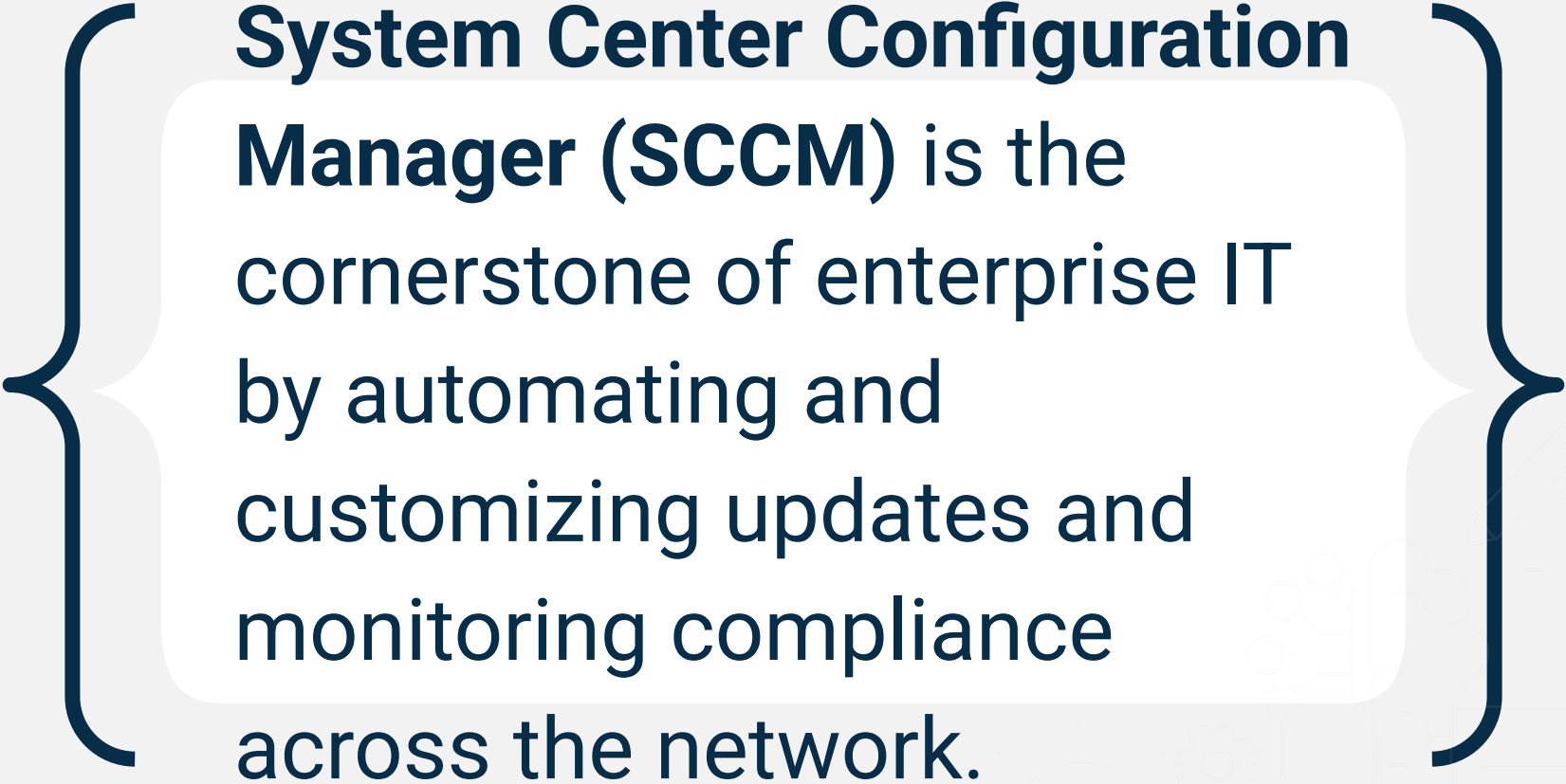
Common reasons for ignoring updates

- **Time** - Systems are expected to up continuously, but patches often require rebooting.
- **Compatibility** - Many services on a Windows server are highly-dependent on other sources. Updating one can affect others.
- **Broken Patches** - Even though patches are tested extensively, there are many variations in hardware and software that can cause issues that may be unnoticed until the patches go to production machines.
- **Limited Resources** - Even if the patch is critical, if the system currently works, techs might delay its deployment.



SCCM and **BCM** to the
rescue!





System Center Configuration Manager (SCCM) is the cornerstone of enterprise IT by automating and customizing updates and monitoring compliance across the network.



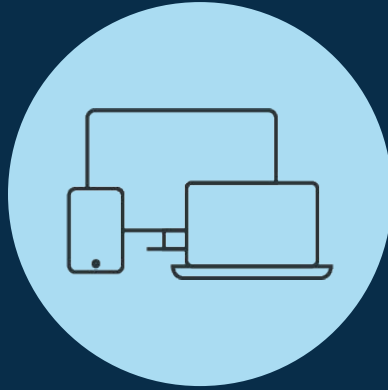
**Business Continuity
Management (BCM)** focuses
on maintaining operations
during disruptions

Patch Tuesday

Occurring on the second Tuesday of each month, Microsoft releases patches for vulnerability and performance enhancements.

While this schedule provides predictability, allowing for organizations to prepare in advance, **Patch Tuesday** also underscores the importance of staying current.





Instructor Demonstration

Configuring Windows Updates



Managing Updates with SCCM

- 1 Create a new Software Update Group
- 2 Select updates from the catalog aligned with compliance policies
- 3 Deploy updates to test environment
- 4 Schedule deployment to live environment
- 5 Monitor compliance using SCCM reporting features



Activity:

Automatically Applying Patches

Suggested Time:
15 Minutes



Time's up!
Let's review



Questions?



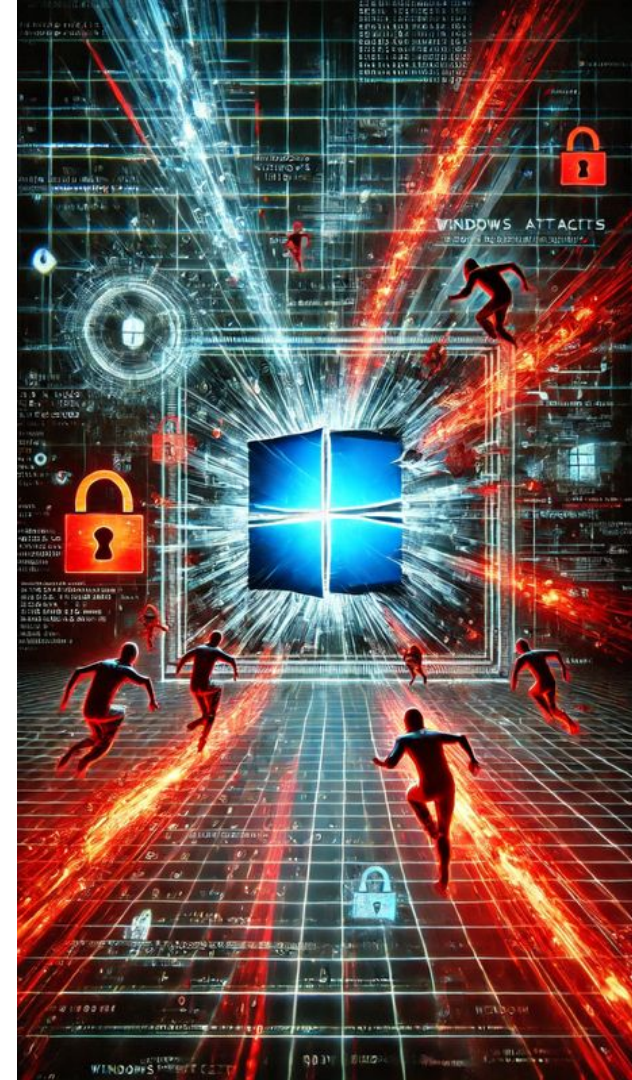


Break
15 mins

Understanding Windows Attack Vectors

Windows Attack Vectors

- **Windows** is a target-rich environment for attackers.
- **Users**, by necessity, often have access to critical files
- **Attackers** exploit this by blending in with legitimate processes, hiding in common locations, and leveraging elevated privileges



Regular User with Local Admin



It starts with a phishing email, but that can lead to opening the floodgates to a full-blown breach!

Attackers rely on error or oversight

- Embedding payloads in temp folders
- Scheduled tasks
- Hidden AppData directories

By blending in with the environment, malware remains undetected

Malware Developers

Over time, developers have become adept at disguising their creations by naming malicious processes to blend in.



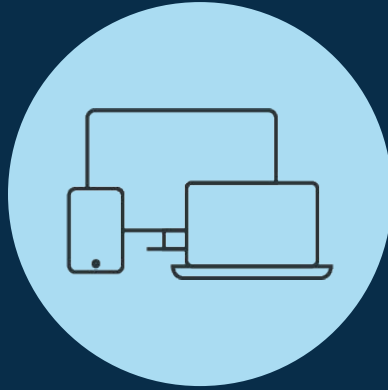


User Space Attacks

Imagine a scenario where a seemingly legitimate service is consuming unusually high CPU cycles.

At first glance, the process name suggests that it is a system service

Let's uncover this hidden threat.



Instructor Demonstration

User Space Attacks



Activity:

Investigating User Spaces and Services

Suggested Time:
20 Minutes





Time's up!
Let's review



Questions?





Thank you